Response to Reviewers

Manuscript: Graph-Based Reinforcement Learning for Electric Truck Fleet Routing and Charging under Operational Uncertainty

We thank the editor and both reviewers for reviews that were, in several places, more accurate about our manuscript than we were. Three of the criticisms turned out to identify real defects rather than presentational weaknesses, and addressing them changed our own conclusions. We state those three at the outset, because they alter claims that appeared in the original submission and we would rather they be visible than discovered.

1. **The feasibility mask does not explain our results.** Prompted by Reviewer 1's comment 2, we implemented a training path without the hard mask. Across three training seeds per arm, the masked and unmasked variants reach mean success rates of 0.760 and 0.773 with overlapping ranges. The mask is a convenience, not the source of the reported behaviour. What is necessary is that infeasibility carry a cost commensurate with failure. The revised manuscript says so in Section 5.6 and no longer attributes performance to masking.

2. **Our optimization benchmark was defective, and is now stronger.** Prompted by Reviewer 1's comment 4 and Reviewer 2's comment 7, we validated the nominal planner against exhaustive enumeration. On tiny deterministic instances brute force beat our "optimal" planner on six of six scenarios, because its circuit formulation could not leave a truck idle and therefore forced every vehicle to serve at least one customer. After correction the planner matches exhaustive enumeration on 30 of 30 instances, and every comparison in the revised manuscript is against the corrected, stronger baseline.

3. **Training-seed variance exceeds most of the effects we previously reported.** Replicating the identical configuration across seeds gives a spread of 0.107 in success rate. Several single-seed comparisons in the original submission are inside that band and are no longer presented as findings. All learned results now report either a seed range or an explicit single-seed marker.

Every experimental claim below is reproducible from artifacts published with the revision, and each response states what changed, where, on what evidence, and what remains open.

Handling Editor

E1. The paper's claim that eVRP addresses single vehicles while eTFRP addresses fleets is incorrect; there is extensive fleet-level eVRP literature, including RL methods.

Response. We accept this without reservation; the dichotomy was wrong as written. The introduction no longer contrasts eTFRP with eVRP on the basis of fleet size.

We have added a matrix (Table 2 of the revised manuscript) recording, for the works this paper already cites, whether each addresses a fleet rather than a single vehicle, whether charging infrastructure is shared, and whether the method is learning-based. Compiling it made the editor's point sharper than we had appreciated: our own bibliography already contained a paper titled "The electric vehicle routing problem with shared charging stations", a heterogeneous-fleet eVRP with nonlinear charging, and two fleet-scale reinforcement-learning treatments. The claim we made was contradicted by work we were citing.

The revised text therefore states that neither the fleet setting, nor shared chargers, nor the use of reinforcement learning is claimed as novel. What is claimed is the specific combination: finite-port queueing whose delay is endogenous to the fleet's own decisions, a nonlinear charging function integrated along the state-of-charge trajectory, correlated travel and energy uncertainty, and event-driven centralized control — supported by the component-level evidence rather than by assertion.

One gap remains and we flag it rather than paper over it. The matrix is built from the works already cited here, which cover electric freight, fleet-level eVRP, shared charging, and fleet-scale RL, but contain no electric bus or ride-sharing exemplars. Those are the two categories the editor names that we cannot populate from our current bibliography, and we will add primary sources for them in the final version.

E2. Operational uncertainty is well studied and the paper presents it as an unfilled gap.

Response. Accepted. The revision separates exogenous randomness (travel time, energy draw, service time) from endogenous delay (queueing, which arises from the fleet's own prior decisions), states the clipping bounds numerically, and presents the contribution as the combination rather than as the treatment of uncertainty per se. Table 2 now records for each quantity whether it is drawn exogenously or arises endogenously.

We also report what this analysis exposed about our own evaluation. In the joint fleet setting of the revised manuscript only two trucks are distributed over twenty-five stations, so charger contention rarely binds, and an ablation that blanks the entire charging-queue state leaves performance inside the seed band. We therefore make no queue-related claim from that setting.

We then constructed a regime in which contention does bind, with four trucks and a single port per station. Mean queueing time rises from 0.23 to 1.92 hours for the CP-SAT planner, confirming that the mechanism is exercised, and the learned policy incurs 1.31 hours against the planner's 1.92, i.e. 32% less waiting. Queue-aware routing is therefore real and measurable; it simply is not what limits performance at small fleet sizes, where the time budget is slack. The main eTFRP benchmark, with fleets an order of magnitude larger over the same twenty-five stations, is the regime in which contention is severe, and we make no new claim about it here because those experiments were not re-run for this revision.

E3. The methodological novelty is unclear; component-level evidence is needed.

Response. The contributions are restated as falsifiable component claims, each tested at matched budget, observation width, curriculum, reward, and seed stream, and re-scored on validation scenarios disjoint from those used for checkpoint selection. Results appear in Section 5.6. Because independently seeded runs of the identical configuration span 0.107 in success rate, we treat effects below roughly 0.1 as indistinguishable from training noise and report them as such.

Three components clear that threshold: the per-action routing features (0.213 when removed, against a 0.700–0.807 seed band), the typed pairwise relations (0.633), and the pooled state embedding (0.667). Two do not: the charging-queue state (0.793) and the active-truck indicator (0.727). We report the negative results alongside the positive ones. On scalability, we no longer claim a computational advantage in the joint setting: the nominal model is small enough that CP-SAT proves optimality in milliseconds, and a budget sweep from 0.5 s to 45 s returns an identical plan.

Reviewer 1

R1.1. In the main eTFRP experiments, deliveries are pre-assigned and mostly ordered, and depot return is not required, so the problem is closer to energy-feasible execution than to fleet routing.

Response. The observation is correct. We have kept the eTFRP as the principal setting because it reflects the operational case this work targets, in which assignments follow from contracts and depot logistics rather than from the routing system, and we now say this explicitly rather than implying that the harder combinatorial problem is being solved.

To address the substance of the criticism, we added a second setting in which precisely the removed decisions are restored: customers are owned by the fleet, assignment and sequencing are decided online, payload capacity binds, and every truck must return to the depot. It replaces the former single-truck eVRP study as the final experimental section, since it answers this comment directly whereas the single-vehicle study did not. The same architecture is applied unchanged and reaches 84.3% success $[83.0, 85.8]$ over three seeds against 62.2% for the corrected CP-SAT planner, while differing from that planner by $+0.2$ hours $[-3.1, +3.7]$ in paired fleet travel time on jointly solved scenarios. See Section 5.5 and Table 7.

R1.2. The feasibility mask may explain most of GraphPPO's gain.

Response. This was the most consequential comment in the review, and the answer is not the one we expected. We implemented a mask-free training path in which the observation and candidate set are identical and only slots denoting no action at all are hidden, so the policy must learn feasibility from experience.

Across three seeds per arm the masked and unmasked variants are indistinguishable: 0.760 against 0.773 mean success, with per-arm ranges of 0.107 and 0.113. The unmasked policy averages 0.1 invalid actions per episode, i.e. it learns feasibility on its own. What proves necessary is the *cost* of infeasibility: when an infeasible action is refused at a tenth of the failure penalty the policy never learns feasibility and success collapses to zero, while at the full failure penalty it recovers to 0.773.

We also compare, under equal information and identical budgets, a flat-state MaskPPO analogue, DeepSets-PPO, a state-GNN with independent action scoring, and a constructive attention model. The action head, not the state encoder, is what separates the family: every independent-head arm lands at 0.527–0.573, while the attention encoder under a complete-GCN head reaches 0.773, inside the graph encoder's seed band. We therefore no longer attribute our results to the graph state encoder either. Reported metrics are feasibility, conditional travel time and makespan, queue time, and per-decision runtime, never reward alone.

R1.3. Notation and equation defects.

Response. All corrected, and the equations were regenerated from the tested implementation rather than edited in place.

- Nominal and realized quantities now use distinct symbols consistently: τ_{uv}, b_{uv} against $\tilde{\tau}_{uv}, \tilde{b}_{uv}$.
- Clipping bounds are given numerically: $\tilde{b}_{uv} \in [0.90 b_{uv}, 1.20 b_{uv}]$ and $\tilde{\tau}_{uv} \in [0.85\tau_{uv}, 2.50\tau_{uv}]$, applied to the realized quantity only, so the nominal network read by the planners is unaffected.
- The charging constraint is rewritten with the state-of-charge trajectory $\mathrm{soc}_i(t + s)$ and elapsed-time integration variable s. We note why it matters: evaluating the power curve at the initial state overstates delivered energy by up to 55% for a session crossing the taper.
- Equation 19 used the receiver embedding twice. Messages now condition on the sender embedding and the edge features, with the receiver entering at the update step.

- Equation 22 summed the node's own embedding over its neighbors. It now aggregates neighboring action embeddings, with self-influence carried by a separate term and the single-feasible-action case defined explicitly.

R1.4. The optimization benchmark does not establish near-optimality.

Response. Accepted, and the investigation prompted by this comment found a defect in our own baseline, reported in the preamble. The inherited per-truck models are now named for what they are in the source and are no longer presented as optimality references. The campaign baseline is a fleet-level CP-SAT model whose objective matches exhaustive enumeration on 30 of 30 tiny instances; an adaptive large neighbourhood search is included as an independent check and reaches the same optimum on all of them. Every episode now publishes solver status, objective, best bound, relative gap, wall time, solve count, and the number of times the executed plan fell back to the shared navigation layer.

We have removed the term "near-optimal" where it was unsupported. A bounded optimality statement is available only on instances small enough to enumerate; at campaign scale we report the best-known plan and say so.

R1.5. High reward is reported despite zero success rate.

Response. Accepted. Reward is no longer a headline metric anywhere. Evaluation is feasibility-first: success rate with Wilson intervals dominates, and cost metrics are reported with their conditioning stated. Every failed episode is retained in every non-conditioned aggregate and is now labelled with a cause; truncation was previously silent, which is where unlabelled failures came from. The inherited shaping has been replaced and every coefficient is published.

R1.6. Baselines are weak and underspecified.

Response. Added: an ALNS metaheuristic with four destroy operators, greedy and regret-2 repair, adaptive operator weights and simulated-annealing acceptance, searching the same nominal costs as CP-SAT and executing through the same energy-safe layer; a constructive attention baseline; DeepSets-PPO; and a state-GNN with independent action scoring. All share the observation, mask, curriculum, reward, seed stream and interaction budget. The claim that PPO is "the most capable discrete-action RL algorithm" has been removed.

Two honest notes. ALNS matches CP-SAT closely, which we take as evidence that the nominal problem is solved rather than that either method is weak. And the attention baseline is competitive with our own encoder, which is why we no longer claim the graph encoder as a contribution.

R1.7. Generalization evidence is too narrow.

Response. The former "zero-shot" study is renamed size transfer, because that is what it measures: only instance size varies. Genuine distribution shift is now studied separately across twenty regimes, each scoring every method on identical held-out scenarios and each labelled interpolation, size transfer, or out-of-distribution so that one kind of evidence is never quoted as another. The main setting is evaluated on 500 paired test scenarios rather than 300, and learned results carry seed ranges.

The finding is bounded rather than uniformly favourable. Transfer holds across instance size and across shifts in charger power, efficiency, port availability, travel and energy variance, demand, and road-network speed, with our method leading thirteen of sixteen regimes. It fails where the resource budget itself moves: with a 300 kWh battery we reach 0.34 against CP-SAT's 0.43, with 500 kWh 0.87 against 0.98, and with doubled service times 0.46 against 0.60. Averaged over out-of-distribution regimes our success changes by -0.110 against -0.006 for ALNS. We state this in the conclusion as a deployment boundary.

Reviewer 2

R2.1–R2.3. The problem definition is confusing; uncertainty and mandatory service should be stated early.

Response. The introduction now defines the joint problem in operational terms without the eT-FRP/eVRP dichotomy, and enumerates the stochastic and endogenous quantities where the problem is introduced rather than later. Every-customer-once service, payload capacity, energy feasibility, the optional time-window variant, and depot return are stated as explicit constraints, and the outcome taxonomy (success, infeasibility, timeout, incomplete service) is defined.

R2.4–R2.6. Language, reward coefficients, charging discretization.

Response. The phrase "stochastic edge traversing" is replaced by "realized travel duration on the selected road-network transition". Every training reward coefficient is published. On discretization, we now have evidence rather than a choice: refining the target state-of-charge grid from 10% to 5% leaves performance inside the seed band, whereas expressing charging as a 15/30/60 minute duration degrades it to 0.627 success and 159.3 travel hours. A fixed duration delivers different energy depending on position on the charging curve, so the same action denotes different outcomes in different states; a target state of charge does not.

R2.7. The optimization model is unclear.

Response. A complete formulation is given with its queueing, uncertainty, horizon, information and fallback assumptions stated, together with which baselines are offline, rolling-horizon, deterministic or scenario-based. Tiny exact objectives are validated against exhaustive enumeration, which is how we found the defect described in the preamble.

R2.8–R2.9. Heuristic and neural baselines; attention and transformer methods should be compared.

Response. A constructive attention baseline is included, adapted to this setting and with the adaptation stated rather than glossed: Kool et al. assume a single vehicle, no charging, and no exogenous uncertainty, so a literal port is impossible; what transfers is the architecture, and we give the typed edge features to it as an attention bias so that it reads exactly the same content as our encoder. Each implemented heuristic is linked to the closest cited variant with its adaptations disclosed.

Minor comments.

Response. "demonstrated" is now "demonstrates"; the missing comma in the objective equation is added; figure caption capitalization is standardized; Table 2 reports realized edge energy rather than only the coefficient ξ. The architecture figure's third panel read "c. Actor Network head" while its second read "b. Value Network Head"; the third is now capitalized to match, so the three panels read "a.", "b." and "c." consistently.

Additional comments

Unloading/service time as a model parameter; endogenous queue delay; sourcing and validating the charging equation; comparison with a three-segment piecewise formulation; an expanded random-variable table.

Response. Service time is defined as a model parameter, sampled once per service event, with its distribution and cap in Table 2. Queue delay is explained as endogenous under finite ports and FCFS admission, with the honest caveat under E2 that the evaluated fleet-to-station ratio does not exercise it.

On the charging function we now report a quantitative comparison rather than an assertion. Against the integrated CCCV curve at 350–750 kW, a constant-power model errs by 5.0% on average and 28.7% at worst; the classical three-segment piecewise-linear formulation of the Montoya type errs by 4.7% and 27.8%, i.e. it is barely better. The reason is structural and we state it: our curve ramps to peak power before tapering and is therefore not concave, while a Montoya-style approximation assumes concavity. Placing breakpoints inside both curved regions reduces the error to 0.6% mean and 4.0% maximum. We also fixed a defect found during this comparison, in which the charge-time estimator failed to converge for a full charge and returned 10 hours for a session requiring 0.53 hours.

Table 2 is expanded to give, for each random variable, its distribution, parameters, clipping bounds, correlation structure, and whether it is exogenous or endogenous.

Do these findings hold in the eTFRP setting?

Because the eTFRP is the principal benchmark, we tested the revision's central finding there rather than only in the joint setting. A pre-assigned variant of the joint formulation was implemented, in which customers are bound to trucks before the episode begins exactly as the eTFRP does, so that the identical architecture, observation, mask machinery and evaluation contract apply to both. Across two seeds per arm, the masked and unmasked policies reach 0.340 and 0.330 mean success against per-arm seed ranges of 0.054 and 0.034. The mask finding therefore replicates in the setting that matters most.

This changes how the manuscript's own PPO-versus-MaskPPO figures should be read. Those two baselines differ in their action representation as well as in masking, so the gap between them is evidence for the structured action space rather than for the feasibility mask specifically, and the revised text now says so instead of attributing it to masking.

A second result from that variant runs against the natural reading of Reviewer 1's first comment. At an identical budget the same model reaches 0.70–0.81 success when it also decides the assignment and only 0.31–0.37 when the assignment is fixed. Removing that decision made the problem *harder*, because a truck must then serve its own customers wherever they fall and the fleet can no longer rebalance after an adverse draw. The eTFRP does ask less of the method combinatorially, as the reviewer says, but it asks more of it operationally.

We are explicit about what was *not* done. The published eTFRP tables were not recomputed: their headline metric is normalized reward, which we now argue against, and their fleet sizes reach an order of magnitude beyond anything scored here. The per-truck Gurobi model behind their optimization column has, however, been examined directly, in two ways.

First, its expressiveness. That model cannot suffer the specific defect we found in the fleet planner, since a fixed assignment leaves no idle-truck decision to get wrong. But the fleet defect was not a solver bug – the model could not express the optimal plan – and that class of defect is testable without a solver. The per-truck model documents the restriction "at most one charger visit between consecutive deliveries". Enumerating tiny deterministic instances, we find that this restriction binds on one of sixty plans, and in that case it turns a route that is feasible in 31.5 hours into one the model reports as infeasible.

Second, its encoding. We solved the model and compared it against exhaustive enumeration over every delivery order and every admissible charger choice, pricing both with the simulator's own nonlinear charging integrator. On every one of the 11 instances (of 80 attempted) where the model returns an executable plan, that plan matches the enumerated optimum exactly. The encoding is therefore sound. The same experiment shows that 42 of 80 of its plans cannot be executed under the simulator's charging physics at all, and that in 7 of those a feasible plan existed under the model's own restriction — the cost of its linear charging curve and flat energy safety factor meeting a nonlinear battery.

The model is thus optimal for a problem slightly different from the one being simulated. We describe that baseline with these qualifications attached rather than as an optimality reference.

Remaining limitations

We prefer to name these rather than leave them to be found.

- Charger contention is modelled but not exercised by the evaluated instances, so no claim about queue-aware behaviour is supported by our results.
- Optimality is bounded only on instances small enough to enumerate.
- Out-of-distribution performance degrades where the energy or time budget moves, and in those regimes a re-solving planner is the better choice.
- The published eTFRP tables retain the defects identified elsewhere in this revision: single-seed comparisons, and an optimization baseline never validated against enumeration.
- Learned results use three training seeds in the joint setting and two in the pre-assigned variant. Given the measured variance we consider three sufficient for the final configuration, whose spread is 0.026, but not for the intermediate training stages, whose spread is 0.120 and whose rankings we therefore no longer interpret.